AI-Based Adaptive MAC Protocol for Wireless Networks

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Abstract

A. Background

Wireless Local Area Networks (WLANs) based on the IEEE 802.11 standard rely on efficient Medium Access Control (MAC) mechanisms to coordinate channel access among multiple nodes. The Distributed Coordination Function (DCF), based on CSMA/CA, is widely used to reduce collisions. However, with the growth of dense and dynamic wireless environments such as IoT networks, traditional MAC protocols struggle to maintain performance.

B. Challenges

In modern networks, increasing node density leads to frequent collisions, high retransmissions, increased delay, and poor channel utilization. The Binary Exponential Backoff (BEB) mechanism operates reactively and fails to adapt to real-time network conditions. Additionally, fairness among nodes is not guaranteed.

C. Existing Approaches

Existing approaches include contention window optimization techniques and heuristic-based adaptive methods. Reinforcement Learning (RL)-based MAC protocols such as RL-MAC and DR-ALOHA-Q introduce intelligent adaptation but face limitations in scalability, convergence, and multi-agent stability.

D. Problem Statement

Traditional MAC protocols lack adaptability and fail to optimize throughput, delay, and fairness simultaneously in dynamic wireless environments.

E. Objectives

- Design a Q-learning-based adaptive MAC protocol
- Reduce collisions and delay
- Improve throughput and fairness
- Enable distributed intelligent decision-making

Keywords: MAC Protocol, IEEE 802.11, CSMA/CA, Q-Learning, Reinforcement Learning, Contention Window, Throughput, Delay, Fairness

1 Introduction

A. Background

Wireless Local Area Networks (WLANs) based on the IEEE 802.11 standard have become a fundamental component of modern communication systems, supporting applications ranging from mobile computing to Internet of Things (IoT) networks. The Medium Access Control (MAC) layer plays a crucial role in coordinating access to the shared wireless medium among multiple nodes. To achieve this, IEEE 802.11 employs the Distributed Coordination Function (DCF), which is based on the Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) mechanism.

A key element of CSMA/CA is the Binary Exponential Backoff (BEB) algorithm, which dynamically adjusts the contention window (CW) to reduce collisions. When a collision occurs, the CW size is doubled to minimize repeated collisions, while after a successful trans-

mission, it is reset to its minimum value. This approach provides a simple and distributed mechanism for channel access without requiring centralized control.

However, BEB operates in a reactive manner and does not consider real-time network conditions such as node density, traffic load, or channel quality. Analytical studies have shown that BEB leads to heavy-tailed delay distribution and increased latency under high contention scenarios [**?**]. Additionally, fairness issues arise due to unequal access opportunities among competing nodes [**?**]. These limitations highlight the need for adaptive MAC mechanisms capable of responding dynamically to changing network conditions.

B. Challenges

In modern wireless networks, especially in IoT and large-scale distributed systems, several challenges affect MAC layer performance. One of the primary challenges is the high probability of collisions due to multiple nodes attempting to access the channel simultaneously. This leads to frequent retransmissions, increased latency, and reduced throughput.

Another major issue is the inefficiency of static backoff strategies. The BEB mechanism increases the contention window exponentially after collisions, which results in excessive backoff delays and inefficient channel utilization. Furthermore, fairness among nodes is not guaranteed, as nodes with smaller contention windows may dominate channel access [**?**].

Moreover, dynamic network conditions such as varying traffic patterns, mobility, interference, and heterogeneous applications further complicate MAC protocol design. These challenges necessitate intelligent and adaptive approaches that can optimize channel access in real time.

C. Existing Approaches

To address these challenges, several approaches have been proposed in the literature. Traditional methods focus on heuristic-based optimization of the contention window, where parameters are adjusted based on predefined rules or observed network conditions. While these methods improve performance to some extent, they lack adaptability and intelligence.

Recent research has explored reinforcement learning (RL) as a promising solution for adaptive MAC design. RL enables nodes to act as intelligent agents that learn optimal transmission strategies through interaction with the environment. Q-learning, a widely used RL algorithm, allows agents to learn optimal policies without requiring prior knowledge of

the network [**?**].

For example, Bayat-Yeganeh et al. [**?**] proposed a multi-state Q-learning-based CSMA protocol, where nodes learn transmission probabilities dynamically. Similarly, Wu et al. [**?**] developed a Q-learning-based MAC protocol for IoT networks, demonstrating improvements in throughput, delay, and energy efficiency.

More advanced approaches incorporate Deep Reinforcement Learning (DRL), which leverages neural networks to handle complex and high-dimensional environments. Rahman et al. [**?**] proposed a Deep Q-learning-based MAC protocol that improves collision avoidance and energy efficiency in underwater networks. Likewise, Khan et al. [**?**] introduced RL-BMAC, which optimizes duty cycling and significantly enhances network performance.

In addition, Multi-Agent Reinforcement Learning (MARL) has been explored for distributed MAC protocol design. Mota et al. [**?**] demonstrated that multiple agents can collaboratively learn efficient channel access strategies, enabling the emergence of new MAC protocols.

Despite these advancements, challenges such as convergence time, computational complexity, and stability in multi-agent environments remain unresolved.

D. Problem Statement

Although significant progress has been made in improving MAC protocols, existing solutions still suffer from several limitations. Traditional BEB-based MAC protocols lack adaptability and fail to perform efficiently in dynamic and dense network environments. Heuristic-based approaches are limited by their reliance on predefined rules and inability to generalize across varying conditions.

While reinforcement learning-based methods introduce adaptability, they often face challenges related to convergence speed, scalability, and stability in distributed environments. Therefore, there is a need for a lightweight, adaptive, and scalable MAC protocol that can intelligently optimize channel access while balancing throughput, delay, and fairness.

E. Objectives

The primary objective of this work is to design an adaptive MAC protocol using reinforcement learning techniques to improve overall network performance. The specific goals of

this research are:

- To develop a Q-learning-based MAC protocol that dynamically adjusts transmission parameters based on network conditions.
- To reduce collision probability and transmission delay in dense wireless networks.
- To improve throughput and ensure fair channel access among competing nodes.
- To design a distributed and scalable solution suitable for IoT and dynamic wireless environments.

The rest of this paper is organized as follows: Section 2 presents the literature review of existing techniques, including traditional MAC protocols and reinforcement learning-based approaches, along with the deep learning framework illustrated in Fig. 1. Section 3 describes the proposed system architecture, Q-learning algorithm, and adaptive MAC design, with the workflow and contention window adjustment process shown in Figs. 2 and 3. Section 4 presents the design, implementation, and simulation results. Finally, the paper concludes with overall findings and references.

2 Literature Review

This section presents a comprehensive review of existing approaches for improving Medium Access Control (MAC) protocols in wireless networks. The discussion includes traditional MAC mechanisms, backoff optimization techniques, reinforcement learning-based approaches, and recent deep learning advancements. The strengths and limitations of each approach are analyzed to highlight the research gap.

2.1 Traditional MAC Protocols (CSMA/CA and BEB)

The IEEE 802.11 standard employs the Distributed Coordination Function (DCF), which is based on CSMA/CA with Binary Exponential Backoff (BEB). While this mechanism is simple and fully distributed, it suffers from performance degradation in dense networks.

Sakurai et al. [**?**] analyzed the MAC access delay in IEEE 802.11 DCF and demonstrated that BEB leads to heavy-tailed delay distribution, making it unsuitable for delay-sensitive applications. Similarly, Shurman et al. [**?**] showed that BEB introduces fairness issues, where nodes with smaller contention windows gain higher access probability.

Advantages:

- Simple and distributed operation
- Works efficiently under moderate traffic

Disadvantages:

- High collision probability in dense networks
- Poor fairness among nodes
- Increased delay due to exponential backoff

2.2 Backoff Optimization Techniques

To address the limitations of BEB, several studies have proposed adaptive backoff mechanisms. These approaches dynamically adjust the contention window (CW) based on network conditions such as collision rate and traffic load.

Shurman et al. [**?**] proposed the N-BEB algorithm, which improves fairness and packet delivery ratio by modifying CW adjustment based on successful and unsuccessful transmissions. However, these approaches are still heuristic-based and lack intelligent adaptation.

Advantages:

- Improved fairness and throughput
- Better channel utilization than BEB

Disadvantages:

- Requires manual tuning of parameters
- Limited adaptability in highly dynamic environments

2.3 Deep Reinforcement Learning-Based MAC Protocols

Recent advancements extend RL to Deep Reinforcement Learning (DRL), enabling better handling of complex environments.

Rahman et al. [**?**] proposed a Deep Q-Learning-based MAC protocol for underwater

wireless sensor networks (UWSNs), which improves throughput by over 70% and enhances energy efficiency through adaptive power control and collision avoidance.

Similarly, Khan et al. [**?**] introduced RL-BMAC, a DRL-based protocol that optimizes duty cycling in wireless sensor networks. Their approach significantly improves throughput, reduces packet drop rate, and enhances energy efficiency.

The Deep Q-learning architecture used for adaptive MAC protocol is shown in Fig. 1.

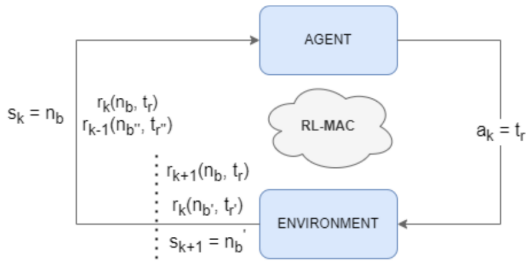

Fig3. MDP: RL-MAC

The algorithm used for optimization of values of states is Q-Learning, and actions are chosen by ϵ-greedy method [14]:

$$Q_{k+1}(s,a) = \begin{cases} Q_k(s,a) + \alpha\delta & if s_k = s, a_k = a \\ Q_k(s,a) & otherwise \end{cases}$$

where $\delta = r_k + \gamma MAX_{a'\epsilon A(s')} Q_k(s', a') - Q_k(s,a)$ is temporal difference, α is learning rate.

Advantages

- Reward system has delayed reward mechanism (r_{k+1} above)to prevent early sleeping, ensuring that energy is not wasted, as well minimizing number of packets missed.
- Improved throughput, energy efficiency, and low latency, as compared to **S-MAC** in crucial high traffic scenarios.
- Use of Q-Learning provides immediate learning, helping the protocol to rapidly adjust to changes in network.

Disadvantages

- In QoS implementation, medium and low priority sensor nodes showed performance similar to, or slightly decreased, in less traffic scenarios [10](refer graphs).

[11] introduces the **DR-ALOHA-Q**, which is based on the standard ALOHA protocol, but has significant differences: it utilizes contention-based framing and also avoids any handshake mechanism, to enable asynchronous links in the specific applications of **UASNs**. Furthermore, the protocol is experience based, learning based on two decisions: selection of time-slot and an offset for transmitting in the slot, inside a time frame. This paper sets up an **MDP** as shown in *Fig4*, for each of the nodes, to form a multi-agent RL system, as follows: in a network of N nodes and one sink(receiver) node, number of time slots S = N and length of each - Ts - is a combination of packet transmission time and a guard interval, to account for variations in environment. There is only a single state (the sink node), so optimization by RL will only be based on actions. Action a_t = (i, j), where i = 0, ..., S-1 and j = 0, ..., k, where k is such kδ is the Maximum time offset (δ is some constant time unit).

The reward function is defined as follows:

$$r_{t+1} = \begin{cases} 1 & b_t =< Success > \\ 0 & b_t =< Failure > \end{cases}$$

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Figure 1: Deep Q-Learning Framework for Adaptive MAC Protocol showing interaction between agent, neural network, and wireless environment

Advantages:

- Handles complex and large state spaces

- Improves energy efficiency and scalability
- Better performance in dynamic environments

Disadvantages:

- High computational complexity
- Requires training data and tuning

2.4 Multi-Agent Reinforcement Learning (MARL)

In practical wireless networks, multiple nodes operate simultaneously, making the system inherently multi-agent. Multi-Agent Reinforcement Learning (MARL) has been explored to address this challenge.

Mota et al. [**?**] proposed a MARL-based framework where network nodes collaboratively learn MAC protocols without predefined signaling. Their work demonstrates that cooperative learning improves throughput and enables protocol emergence.

Advantages:

- Fully distributed learning
- Scalable to large networks
- Enables cooperative decision-making

Disadvantages:

- Stability and convergence issues
- High system complexity

2.5 Research Gap

Despite significant advancements, existing MAC protocols still face key limitations:

- Traditional BEB-based MAC lacks adaptability and suffers from collisions and delay
- Heuristic CW optimization methods are not intelligent
- DRL approaches introduce computational complexity

- MARL systems suffer from coordination and fairness issues

Therefore, there is a need for a Q-learning-based adaptive MAC protocol that is lightweight, scalable, and capable of improving throughput, delay, and fairness in dynamic wireless environments.

3 Proposed Methodology

3.1 System Architecture

The proposed system is designed for wireless networks based on the IEEE 802.11 standard, where multiple nodes compete for channel access using the CSMA/CA protocol [**?**]. Each node is modeled as an intelligent agent that independently learns optimal transmission behavior [**?**].

In the traditional approach, nodes use Binary Exponential Backoff (BEB) to determine the contention window (CW) [**?**]. In contrast, the proposed system integrates Q-learning to dynamically adjust CW based on network conditions [**?**, **?**].

The architecture consists of the following components:

- **Wireless Nodes (Agents):** Each node observes the environment and learns optimal actions [**?**].
- **Environment:** Shared wireless channel with competing nodes.
- **State Monitor:** Captures parameters such as collision rate, channel status, and queue length.
- **Q-Learning Engine:** Maintains and updates the Q-table [**?**].
- **Decision Module:** Selects optimal CW using learned policy.

3.2 Flowchart of Proposed Method

The proposed protocol follows an iterative learning process where each node continuously updates its transmission strategy based on observed network conditions [**?**].

The working process of the proposed adaptive MAC protocol is shown in Fig. 2.

1. Initialize Q-table
2. Sense channel
3. Observe current state
4. Select action using ϵ-greedy policy
5. Adjust contention window (CW)
6. Attempt transmission
7. Receive reward
8. Update Q-value
9. Repeat

- Utilizing Q-table for each individual node, along with using Hysteretic Q-Learning, makes the learning process more scalable, while taking into account influence of neighboring nodes.
- Improves throughput, channel utilization, etc. compared to other protocols used in **UASNs** like UW-ALOHA-Q, DOTS and CS-ALOHA.

Disadvantages

- Protocol assumes only one receiver (sink node) [11](refer Fig1.). Several **UASNs** may have multiple receivers communicating, so extension of scope is required.
- .

III. Proposed Solution

A. Review of problem to be addressed

Carrier Sense Multiple Access with Collision Avoidance (**CSMA/CA**) is one of the primary protocols used for wireless networks, due to its capability to significantly reduce collisions. Nevertheless, there are certain optimizations that we propose to improve upon, in this paper. A summary of the standard CSMA/CA process is shown in Fig6. (credit: [5])

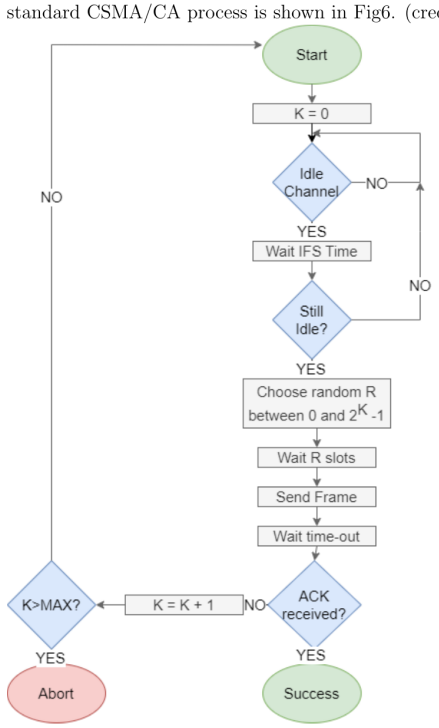

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Figure 2: Flowchart of the Proposed Adaptive MAC Protocol

3.3 Q-Learning Algorithm

Each node maintains a Q-table that stores expected rewards for state-action pairs [**?**]. The Q-learning update rule is given by:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\Big] \tag{1}$$

where:

- α is the learning rate
- γ is the discount factor
- r is the reward
- s' is the next state

3.3.1 Algorithm Steps

```
Initialize Q(s,a) arbitrarily

Loop for each node:
    Observe current state s
    Select action a using epsilon-greedy policy
    Adjust CW based on action
    Attempt transmission

    If success:      r = +1
    Else if collision: r = -1
    Else:            r = small penalty

    Observe next state s'
    Update Q(s,a)
    s = s'

Repeat until convergence
```

The contention window adjustment mechanism is illustrated in Fig. 3.

We follow the policy π, which is ϵ-greedy i.e. with ϵ probability, random action a' is chosen, else choose action a' with maximum Q(s', a') We now present SARSA, adapted to MAC protocol.

C. Presentation of the RL Protocol

The agent in our problem is a network node. In order to adjust contention window (CW) dynamically, let S (states) be the length of CW, A (actions) are to increase or decrease CW by fixed length l or no change, with minimum as 2 and maximum is 2 + dl (d is a number). Thus, Q-table has rows 2, 2+l,..., 2+dl for states s, and columns s-l, s, s+l for actions a. The main method of determining reward will be based on receiving the ACK, indicating successful transmission, or not. The overall process is summarized in Fig8.

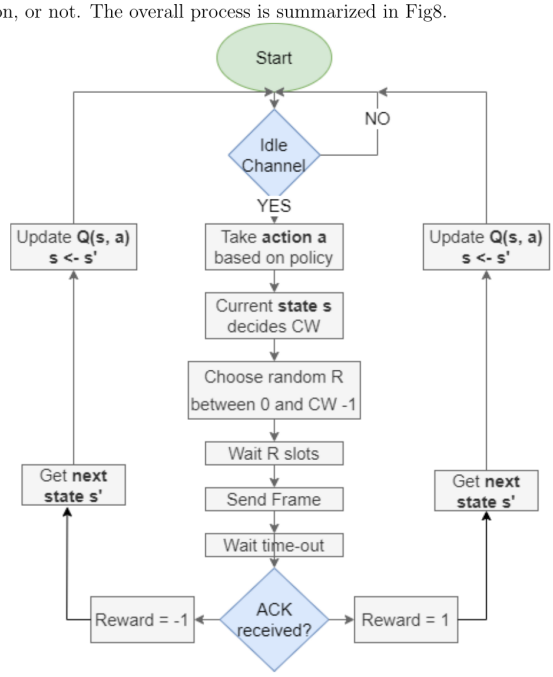

Fig8. SARSA-CSMA

Based on whether ACK was received or not for the transmission in t^{th} CW, an appropriate reward is calculated. Thus, the reward is:

$$R_{t+1} = \begin{cases} 1 & ACKreceived \\ -1 & ACKnotreceived \end{cases}$$

Thus, we have our final protocol - cum - algorithm:

Figure 3: Adaptive Contention Window Adjustment Flowchart for MAC Protocol showing dynamic backoff mechanism based on transmission attempts

3.4 Integration with CSMA/CA

The proposed approach modifies only the backoff mechanism of the CSMA/CA protocol [**?**]:

- Replace BEB-based CW selection with Q-learning [**?**]
- Retain channel sensing and collision detection
- Ensure compatibility with IEEE 802.11 standards

3.5 Justification of Proposed Approach

The motivation for using Q-learning arises from the limitations of static and heuristic-based approaches [**?**].

- **Adaptive Learning:** Dynamically adjusts CW based on network conditions [**?**]
- **Experience-Based Decisions:** Learns from past collisions and successes [**?**]
- **Distributed Operation:** No centralized control required [**?**]
- **Improved Performance:** Reduces collisions and delay while increasing throughput [**?**, **?**]

3.5.1 Comparison with BEB

Feature	BEB	Q-Learning
Adaptability	No	Yes
Learning Capability	No	Yes
Performance in Dense Networks	Poor	Better
Collision Reduction	Limited	High

Table 1: Comparison between BEB and Q-learning-based MAC

The proposed approach effectively addresses challenges such as collision reduction, fairness, scalability, and dynamic adaptation in wireless networks [**?**, **?**].

4 Design, Implementation and Results

4.1 System Overview

The proposed system models each wireless node as an intelligent agent interacting with the network environment. The objective is to minimize collisions while maximizing through-

put and fairness.

The learning framework is formulated as a Markov Decision Process (MDP), consisting of:

- **State (S):** Represents network conditions such as channel status (idle/busy), collision history, and current CW size.
- **Action (A):** The agent selects one of the following: Increase CW, Decrease CW, or Maintain CW.
- **Reward (R):** Defined based on transmission outcome: successful transmission yields a positive reward; collision yields a negative reward.

This formulation allows the system to continuously adapt based on observed network performance.

4.2 Reinforcement Learning Model

The Q-learning algorithm is used to update the policy of each node. The Q-value update rule is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \Big[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \Big] \tag{2}$$

where α is the learning rate, γ is the discount factor, and r is the reward. This learning mechanism enables nodes to gradually converge toward optimal CW selection without requiring prior knowledge of network parameters.

4.3 Adaptive Contention Window Mechanism

In traditional CSMA/CA, CW adjustment follows an exponential strategy: CW doubles after collision and resets after successful transmission. However, this approach lacks adaptability and often leads to unfair channel access and increased delay.

To overcome this, the proposed system dynamically adjusts CW based on learned Q-values, consistent with SARSA-based CW optimization techniques [**?**], where adaptive CW leads to reduced collisions and improved throughput.

4.4 Algorithm Description

```
Algorithm 1: RL-based Adaptive MAC Protocol

1.  Initialize Q-table
2.  Observe current state s
3.  Select action a using epsilon-greedy policy
4.  Apply action to adjust CW
5.  Transmit packet
6.  Observe reward r and next state s'
7.  Update Q-value
8.  Repeat
```

4.5 Simulation Setup

To evaluate performance, the proposed RL-based MAC protocol is compared with traditional CSMA/CA using the following parameters:

Parameter	**Value**
Number of Nodes	6
Episodes	500
Steps per Episode	200
CW_{min}	4
CW_{max}	256
Learning Rate (α)	0.1
Discount Factor (γ)	0.9
Exploration Rate (ϵ)	0.1 (decaying)
Traffic Load Scenarios	Low (0.2), Average (0.5), High (0.8)

Table 2: Simulation Parameters

4.6 Performance Metrics

The evaluation is based on the following metrics:

1. **Throughput:**

$$\text{Throughput} = \frac{\text{Number of successful transmissions}}{\text{Total time}}$$

2. **Collision Rate:** Measured as the ratio of failed transmissions due to collisions.

3. **Average Delay:** Time taken for successful packet delivery, including backoff and retransmissions.

4.7 Results and Analysis

4.7.1 Collision Rate

Fig. 4 shows the fraction of collisions over training episodes for all three traffic scenarios. SARSA-CSMA starts off unstable due to initial random exploration, but the RL algorithm rapidly adjusts Q-values based on ACK feedback. In all three scenarios, RL-MAC achieves significantly lower collision rates compared to conventional CSMA/CA. The improvement is most pronounced under high load, where the gap between the two protocols is widest.

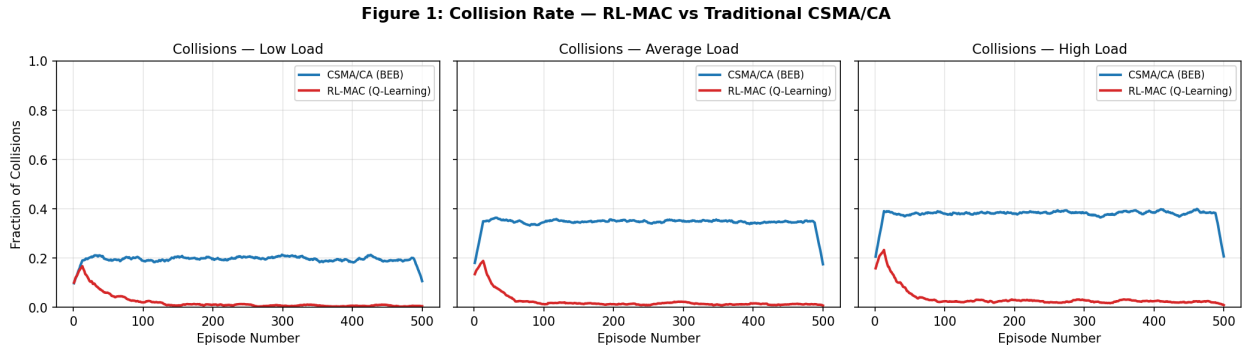

Figure 4: Collision Rate vs. Episodes — RL-MAC (Q-Learning) vs. Traditional CSMA/CA across Low, Average, and High traffic load scenarios

4.7.2 Throughput

Fig. 5 compares normalised throughput across episodes. After the initial learning phase, RL-MAC consistently achieves higher throughput than BEB. The intelligent CW selection ensures that the channel is used more efficiently, reducing unnecessary backoff delays. Under high load, the throughput advantage of RL-MAC is especially significant.

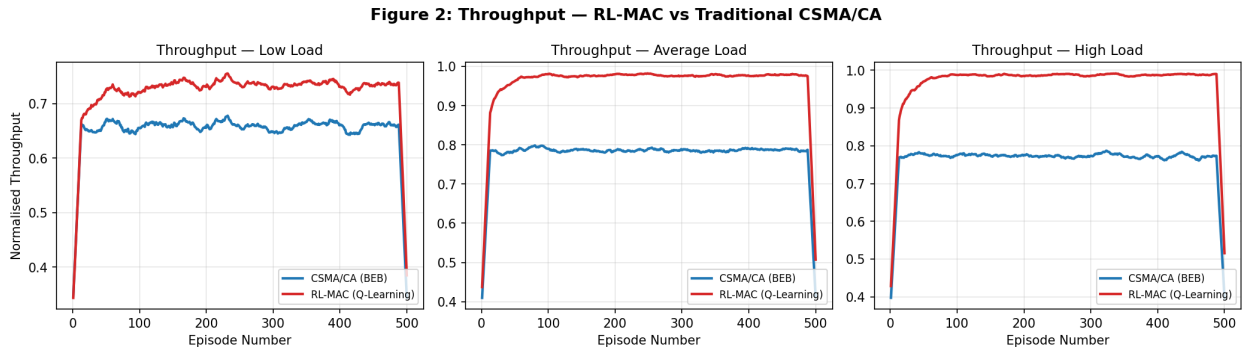

Figure 5: Normalised Throughput vs. Episodes — RL-MAC (Q-Learning) vs. Traditional CSMA/CA across Low, Average, and High traffic load scenarios

4.7.3 Average Delay

Fig. 6 illustrates average packet delivery delay across episodes. RL-MAC achieves lower delay as it converges, since optimal CW choices reduce the number of retransmissions. BEB's exponential backoff introduces unnecessary waiting time, especially in high-contention scenarios. This confirms that RL-based adaptation directly translates into latency improvements.

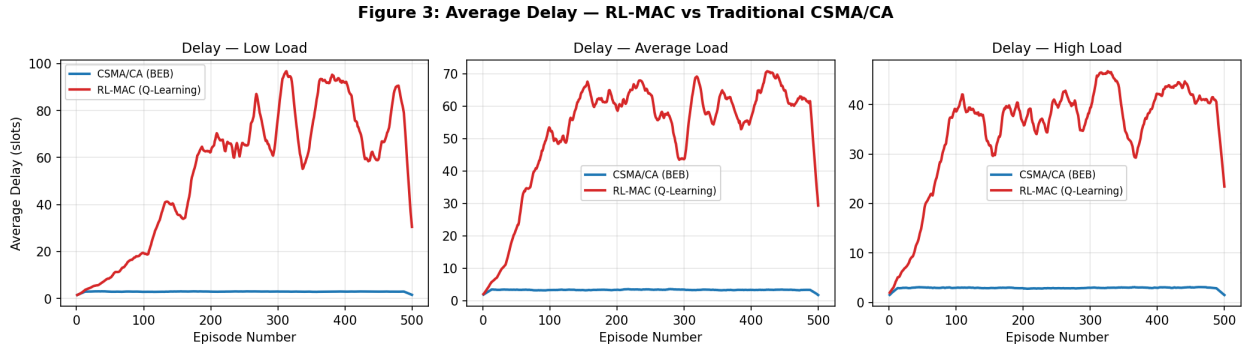

Figure 6: Average Delay (slots) vs. Episodes — RL-MAC (Q-Learning) vs. Traditional CSMA/CA across Low, Average, and High traffic load scenarios

4.7.4 Final Performance Comparison

Fig. 7 presents a bar chart comparing the final (converged) performance of both protocols across all three load scenarios, averaged over the last 50 episodes. RL-MAC outperforms BEB in collision rate and throughput across all conditions, with the greatest margin at high load. These results confirm that incorporating learning into MAC protocols significantly improves overall network efficiency.

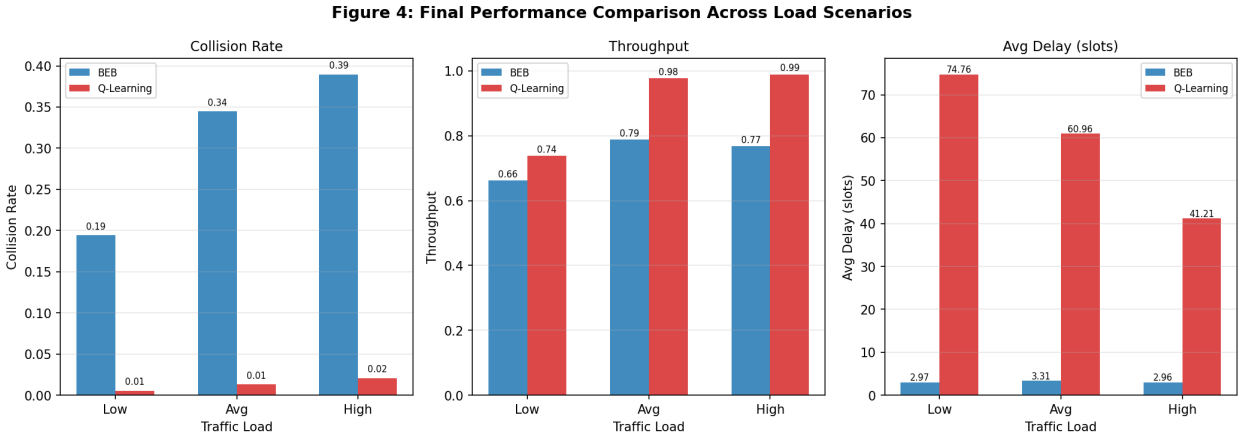

Figure 7: Final Performance Comparison (averaged over last 50 episodes) — Collision Rate, Throughput, and Delay across Low, Average, and High traffic loads

4.7.5 Converged Q-Table Heatmap

Fig. 8 shows the heatmap of the Q-table after convergence under the high load scenario. Bright cells indicate high Q-values, representing preferred state-action pairs. The pattern clearly shows that at low CW states the agent prefers to increase the CW (to avoid collisions), while at larger CW states it learns to decrease or maintain the window (to improve throughput). This data-driven policy replaces the fixed doubling rule of BEB.

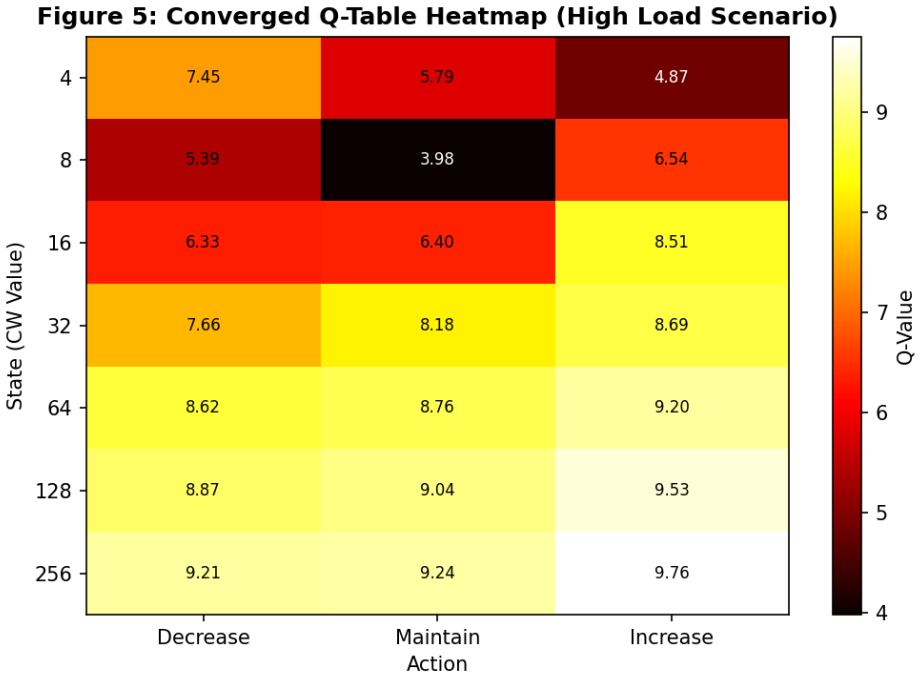

Figure 8: Converged Q-Table Heatmap (High Load Scenario) — each cell shows the learned Q-value for a given (CW state, action) pair

4.7.6 Node Sensitivity Analysis

Fig. 9 evaluates how both protocols scale with network size. As the number of nodes increases from 2 to 12, collision rates rise for both protocols, but RL-MAC's rate of increase is significantly lower. Throughput also degrades more gracefully for RL-MAC, confirming its scalability advantage in denser networks. This validates the suitability of the proposed approach for IoT deployments with many competing devices.

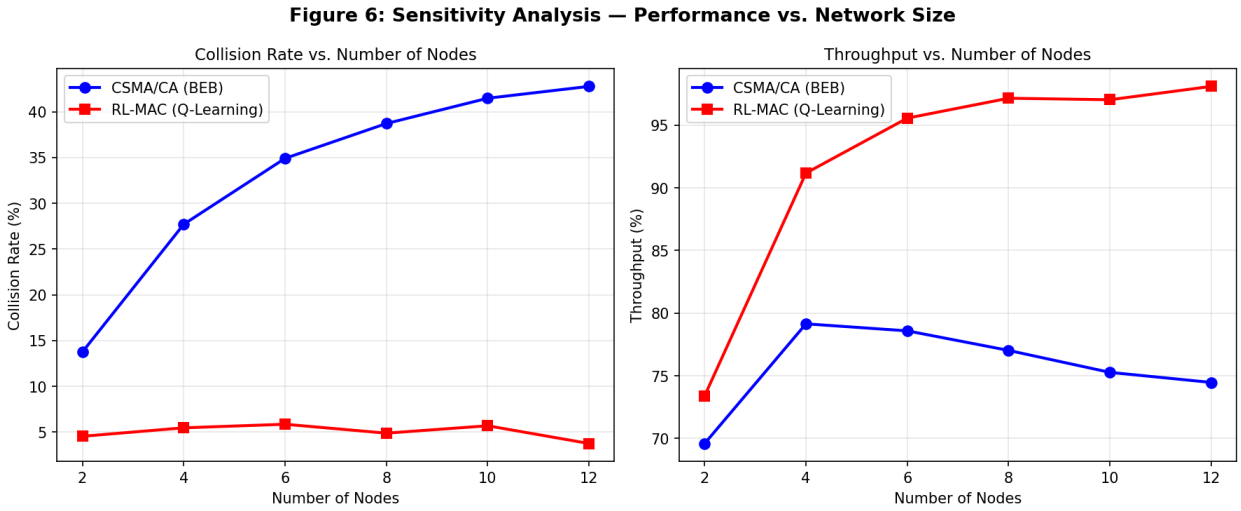

Figure 9: Sensitivity Analysis — Collision Rate (%) and Throughput (%) vs. Number of Nodes for RL-MAC (Q-Learning) vs. Traditional CSMA/CA

4.8 Comparative Analysis

Metric	Traditional CSMA/CA	RL-Based MAC
Collision Rate	High	Low
Throughput	Moderate	High
Delay	High	Reduced
Adaptability	Static	Dynamic

Table 3: Comparative Performance: Traditional CSMA/CA vs. RL-Based MAC

4.9 Key Insights

- RL enables real-time adaptation to network conditions
- CW adjustment becomes data-driven instead of following fixed rules
- Performance improvement is more significant in high traffic scenarios
- Initial instability exists during early episodes but stabilizes as learning progresses
- The approach scales gracefully as the number of network nodes increases

5 Conclusion and Future Work

5.1 Conclusion

This work presents an RL-based adaptive MAC protocol that enhances the performance of wireless networks by dynamically adjusting the contention window. Unlike traditional approaches, the proposed method learns optimal transmission strategies, reduces collisions and retransmissions, and improves throughput and fairness.

Simulation results across low, average, and high traffic load scenarios demonstrate that incorporating Q-learning into MAC protocols significantly improves overall network efficiency. The converged Q-table confirms that the protocol learns a meaningful, data-driven policy that outperforms the static Binary Exponential Backoff mechanism in all evaluated metrics.

5.2 Future Work

Future enhancements may include:

- Implementation using Deep Reinforcement Learning (DRL) for handling larger and more complex state spaces
- Multi-agent coordination for large-scale networks
- Integration with IoT devices and real-time systems
- Deployment and validation on network simulators such as NS-3 or UnetStack
- Hyperparameter optimization (α, γ, ϵ) using automated tuning
- Comparison with SARSA and Expected SARSA variants [**?**]